

TogoCoord

(provisional)

An initiative for converting sequence coordinates across biological layers



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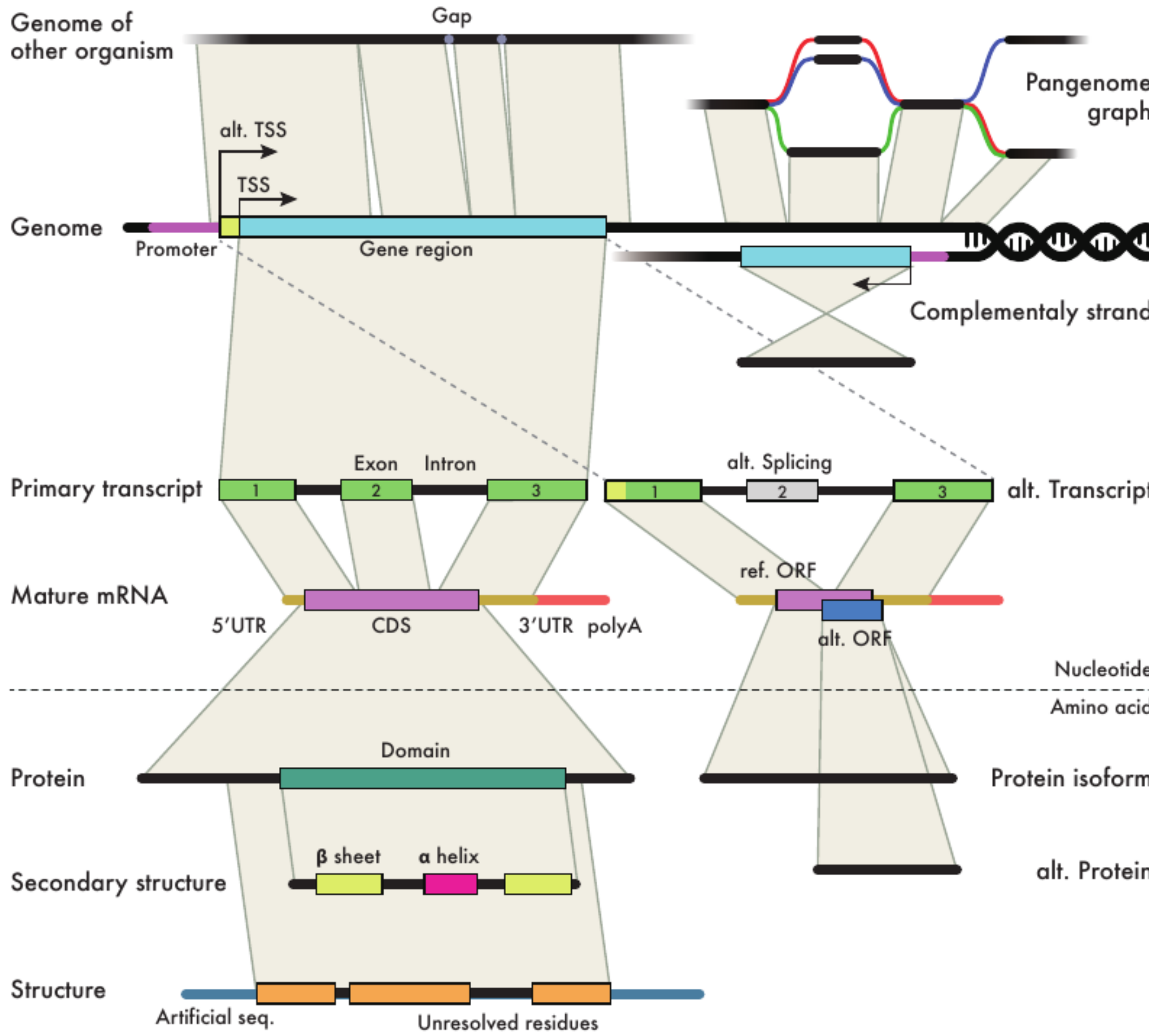
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Sequence information in the life sciences is represented at multiple biological layers, including genomes, transcripts, and proteins, and therefore uses multiple coordinate systems. Representative examples include genomic, pre-mRNA, CDS, and amino-acid coordinates. Conversions among these systems are frequently required in research, for example when mapping a genomic variant to a UniProt protein sequence or a PDB three-dimensional structure, or when comparing variants in orthologs across species.

Range-to-range conversion is especially complex because it must account for discontinuities caused by exon-intron structures, reverse-strand genes, isoform differences, and variants with insertions or deletions. Until now, such cases have largely depended on ad hoc solutions. To address these challenges, we initiated TogoCoord (provisional), a general-purpose, reusable coordinate-conversion service.

Diverse coordinate systems

Sequences at different biological layers are connected through the central dogma and through evolution. By tracing these relationships, corresponding positions can be identified across layers.

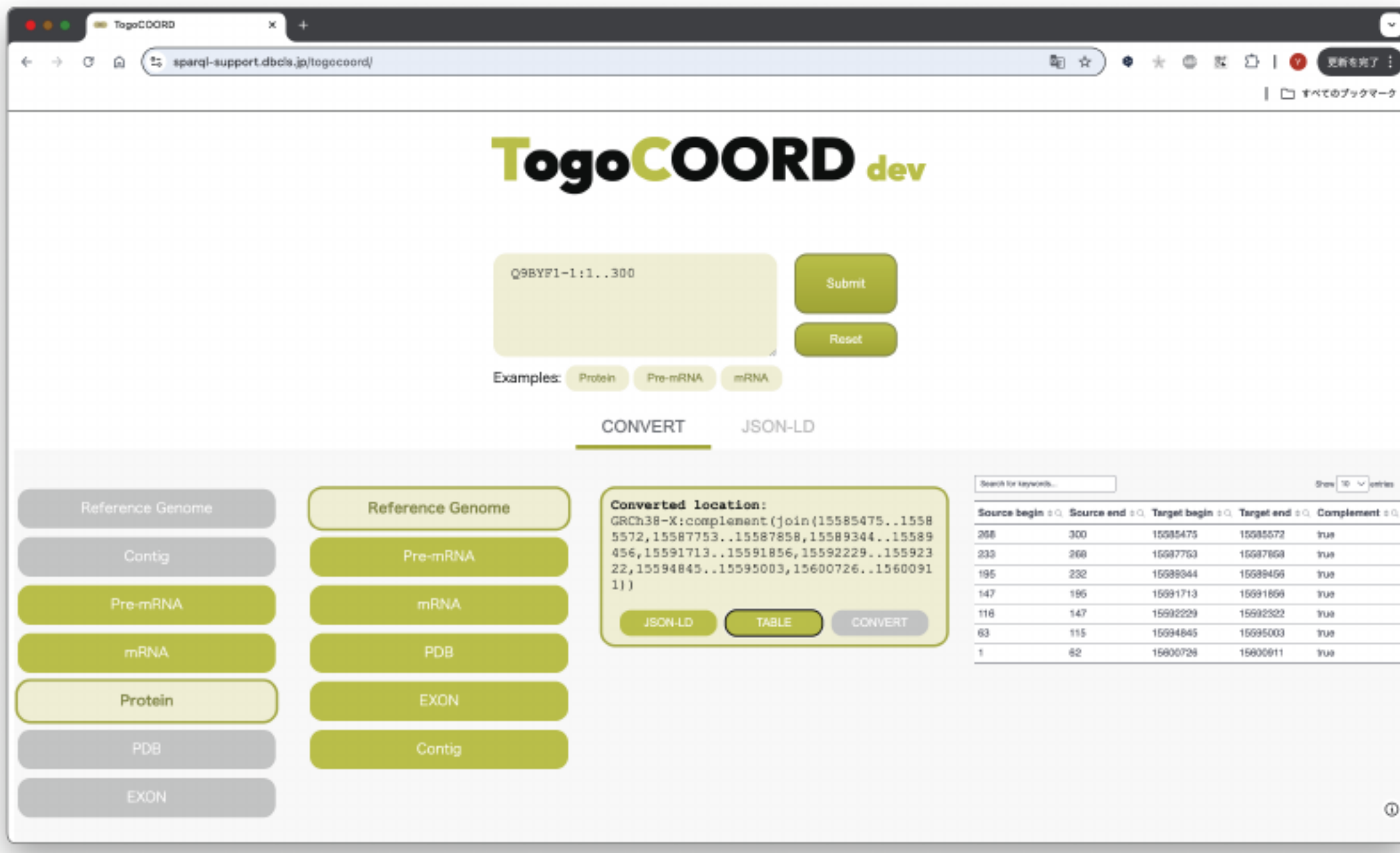


In practice, however, discontinuities, inversions, and inserted sequences increase complexity. Mappings may be discontinuous, and both order and strand orientation may change, making coordinate conversion difficult.

Even within the same biological layer, sequences in different databases are not always identical, so coordinates cannot simply be shared. Even in human, the MANE Select protein sequence may differ from the UniProtKB/Swiss-Prot canonical sequence; approximately 560 MANE sequences instead match a TrEMBL entry.

Implementation and UI

We collected linking data and APIs and built a prototype for human, mouse, and other species. It converts arbitrary positions - annotation coordinates - across biological layers. Positions and compound locations are represented as Location IDs based on INSDC Feature Table location notation.



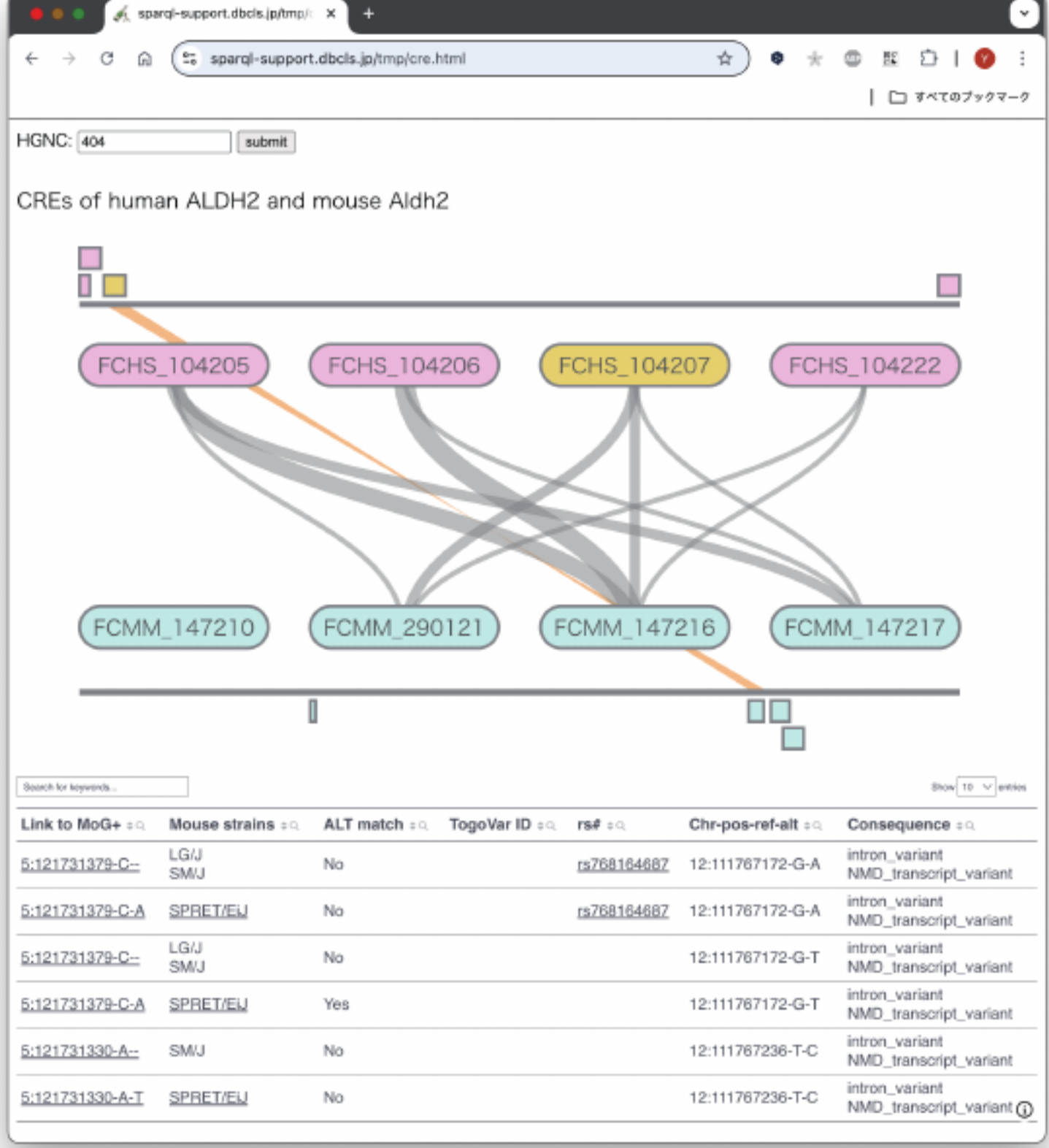
Prototype: converting a protein interval to genomic coordinates

Coordinate conversion propagates annotations

Projecting a position into another layer makes it possible to compare annotations across layers. For organisms with limited data, conversion through a well-annotated species can also unlock a broader range of annotation information.

Human and mouse genomic coordinates

Coordinate conversion with shared ortholog and transcription-factor data enables comparison of variants in corresponding cis-regulatory elements (CREs) between human and mouse, helping identify experimental mouse strains. (fanta.bio, ChIP-Atlas, MoG+, and TogoVar)



Coordinate conversion by liftOver using UCSC chain files

These conversions are not provided through a single, unified mechanism, and each implementation often requires substantial effort. A comprehensive coordinate service such as TogoCoord could prevent repeated reinvention of the same conversion workflows.

Location ID and FALDO JSON-LD

<https://github.com/ddbj/universal-public-genome>

GCA000000000-J00000:467

```
{
  "@context": "http://example.org/context/faldo.jsonld",
  "id": "http://example.org/GCA000000000-J00000-location",
  "faldo:location": {
    "type": "faldo:ExactPosition",
    "faldo:position": 467,
    "faldo:reference": "insdc:J00000"
  }
}
```

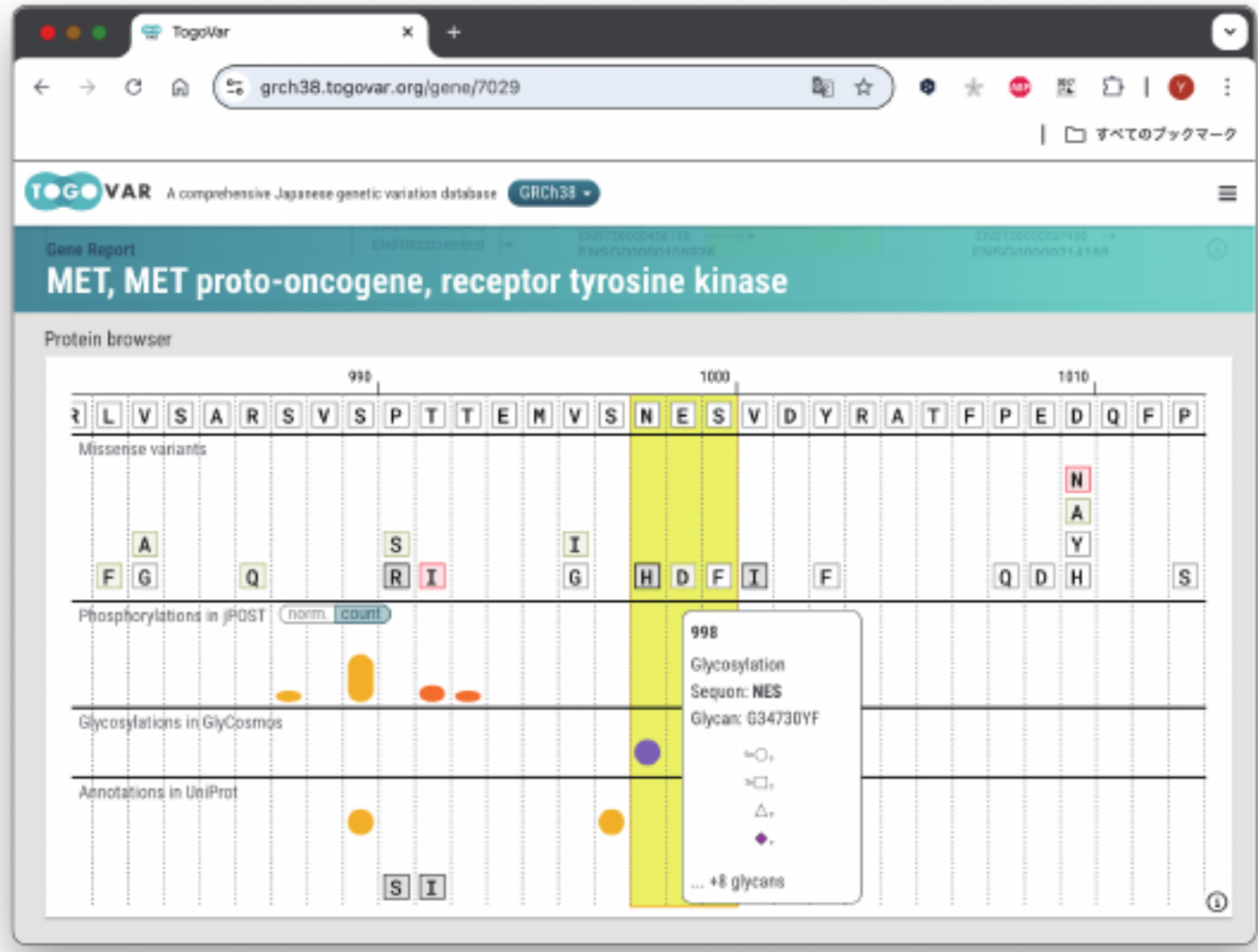
A new URI composed of a sequence ID and position
All sequence-derived annotations can be expressed as Location IDs

GCA000000000-J00000:complement(join(2691..4571,4918..5163))

```
{
  "@context": "http://example.org/context/faldo.jsonld",
  "id": "http://example.org/GCA000000000-J00000-location",
  "faldo:location": {
    "type": "faldo:ListOfRegions",
    "faldo:strand": "faldo:NegativeStrand",
    "faldo:member": [
      {
        "type": "faldo:Region",
        "faldo:begin": {
          "type": "faldo:ExactPosition",
          "faldo:position": 2691,
          "faldo:reference": "insdc:J00000"
        },
        "faldo:end": {
          "type": "faldo:ExactPosition",
          "faldo:position": 4571,
          "faldo:reference": "insdc:J00000"
        },
        "faldo:order": 1
      },
      {
        "type": "faldo:Region",
        "faldo:begin": {
          "type": "faldo:ExactPosition",
          "faldo:position": 4918,
          "faldo:reference": "insdc:J00000"
        },
        "faldo:end": {
          "type": "faldo:ExactPosition",
          "faldo:position": 5163,
          "faldo:reference": "insdc:J00000"
        },
        "faldo:order": 2
      }
    ]
  }
}
```

Genomic and amino-acid coordinates

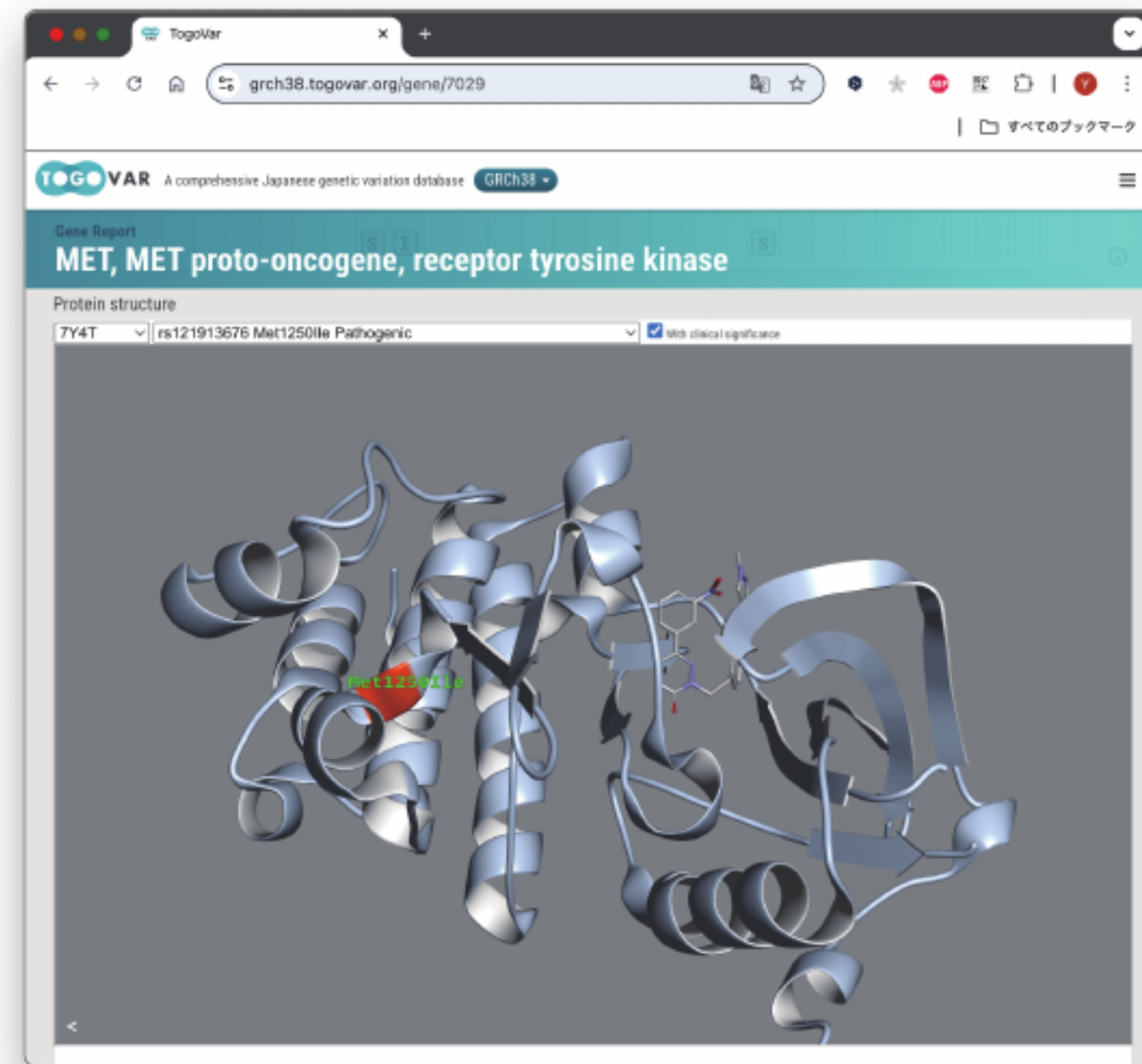
Visualize the amino-acid substitution caused by a variant and its potential effects on phosphorylation or glycosylation sites in the original protein sequence. (TogoVar)



Conversion based on annotations from Ensembl Variant Effect Predictor (VEP)

Genomic and structural coordinates

Map a variant onto a PDB structure and classify the affected residue as surface-exposed or buried. (TogoVar)



Conversion based on PDB-UniProt alignments while excluding low-confidence residues

Outlook

The development version covers selected layers for human and a limited set of other species. For some organisms, obtaining the data required to link coordinate systems - or using public services such as the Ensembl API - remains difficult.

We therefore plan to standardize a minimal set of genomic coordinate conversions and to develop implementations for LiftOver to well-annotated assemblies and closely related species.



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